

PROVENANCE OF THE POLYNOMIAL GLOBALIZATION ARGUMENT IN THE LECH MANUSCRIPT

RESEARCH RETROSPECTIVE PREPARED WITH CODEX

ABSTRACT. This note responds to a reviewer’s question about the origin of the polynomial-resolution construction in the equal-characteristic Lech manuscript. We distinguish elementary descent to the localized polynomial ring from a globally exact polynomial resolution with prescribed local ranks and column orders. The latter construction in the manuscript uses a fixed direct sum of the given module. We record its classical inputs, the documented development on September 4, 2026, and relevant earlier literature. The evidence supports a synthesis of Forster–Swan generator theory, a relative incidence argument, and Quillen–Suslin. It does not establish novelty or identify a particular training source. This is a provenance report, not external certification of the complete Lech argument.

1. THE SHORT ANSWER

The clearest documented answer is that this construction arose to fill a specific gap in the proposed proof: a graded inequality had to be transferred to a local resolution without losing exactness, the local Betti ranks, or the depths of selected columns. The solution was to construct a polynomial resolution inside actual syzygy modules, using direct sums to absorb the dimension cost in a global generator bound. Forster–Swan handled the intermediate syzygies, and Quillen–Suslin handled the final projective syzygy. A prescribed-jet refinement was then added to retain selected deep columns.

The contemporary notes explicitly cite the classical ingredients but do not cite a published source for that entire assembled proposition. A targeted literature search for this report found relevant antecedents, not a verified exact match. Consequently, neither “this was copied from a known theorem” nor “this is a new theorem” is established by the present evidence.

There is also an important qualification to the reviewer’s paraphrase: the global construction is stated for $M^{\oplus a}$, not necessarily for M itself. The weaker statement that the completed minimal resolution can be represented by polynomial matrices is a different, considerably more elementary assertion.

2. THREE STATEMENTS THAT SHOULD BE KEPT SEPARATE

Throughout, let

$$S = k[x_1, \dots, x_d], \quad \mathfrak{m} = (x_1, \dots, x_d), \quad A = k[[x_1, \dots, x_d]], \quad d \geq 1.$$

Let $M \neq 0$ be a finite-length A -module, and let F be its minimal free resolution.

2.1. Descent of the module. Choose C such that $\mathfrak{m}_A^C M = 0$. The canonical isomorphism

$$S/\mathfrak{m}_A^C \simeq A/\mathfrak{m}_A^C \tag{2.1}$$

makes M itself a finite S -module supported at the origin. There is no approximation issue at this stage.

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2.2. Polynomial matrices after localization or completion. Take a minimal free resolution of M over $S_{\mathfrak{m}}$ and complete it. Flatness and preservation of the residue field give a minimal A -resolution, hence a complex isomorphic to F .

There are only finitely many denominators in its matrices. Multiplying each differential by a suitable polynomial outside $\mathfrak{m}$ gives polynomial matrices. Their consecutive products are zero, since they are zero after localization and S injects into $S_{\mathfrak{m}}$. After completion, the scalar multipliers are units and can be absorbed into degree-wise changes of basis. Thus these matrices represent the same minimal completed resolution.

This does *not* by itself give a globally exact S -free resolution of the particular S -module M . Additional homology or support can remain away from the origin. For example, multiplication by $x(1-x)$ over $k[x]$ becomes a minimal presentation of k after completion at (x) , but its global cokernel also has support at $x = 1$. This example illustrates the local/global distinction; it is not a counterexample to the existence of a better global resolution.

2.3. The globally exact assertion in the manuscript. The proposition on “globalization with selected columns” records the following stronger result; see the manuscript and original notes listed in Appendix A.

Proposition 2.1 (Statement recorded in the manuscript). *In each positive degree i , mark g_i basis vectors of F_i whose differential columns have entries in $\mathfrak{m}_A^N$, where $N \geq 1$. If k is infinite, then for every integer $a \geq d$ there is an exact global S -free resolution P of $M^{\oplus a}$, minimal at $\mathfrak{m}$, such that*

$$\text{rank}_S P_i = a\beta_i(F), \quad A \otimes_S P \simeq F^{\oplus a},$$

and at least ag_i columns in degree i have entries in $\mathfrak{m}^N$. If k is finite, the same conclusion holds for some positive integer a .

The earlier all-deep version works for every $a \geq d$ over any field when one asks only to preserve the orders of whole differential matrices. The selected-column refinement is the reason for the extra field argument in Proposition 2.1.

Remark 2.2. The manuscript does not assert the unstabilized global conclusion for M in this proposition. This qualification is a limit of the recorded construction, not evidence that the unstabilized assertion is false. In its application, lengths, Betti ranks and selected-column counts all gain the same factor a , which is subsequently divided out. The polynomial resolution is ungraded; the following periodic-box step produces the graded objects.

3. HOW THE CONSTRUCTION WORKS

3.1. Direct sums create the required generator surplus. Write

$$r_i = \text{rank}_A \text{im}(F_i \longrightarrow F_{i-1}).$$

For the intermediate stages $1 \leq i < d$,

$$\beta_i = r_i + r_{i+1}, \quad r_{i+1} > 0. \tag{3.1}$$

Construct the global resolution of $M^{\oplus a}$ one actual kernel K_i at a time. At the origin it needs $a\beta_i$ generators. Off the origin the preceding exact sequences split locally, so K_i is locally free of rank ar_i .

The Forster–Swan bound yields

$$\mu_S(K_i) \leq \max\{a\beta_i, ar_i + d\} = a\beta_i, \tag{3.2}$$

because

$$a\beta_i - ar_i = ar_{i+1} \geq a \geq d. \tag{3.3}$$

Thus amplification absorbs the dimension penalty while retaining exactly the local minimal ranks. The source consulted at the time for the precise generator bound was [6, Theorem 1.1]. The classical ancestry is Forster–Swan and the basic-element work of Eisenbud–Evans [7, 10, 4, 5].

3.2. Prescribed jets retain selected column depths. The ideal of all entries in a differential is invariant under invertible source and target changes. The orders of particular source columns are not invariant under arbitrary source changes. Ordinary global generation therefore does not settle the selected-column problem.

The refinement lifts desired generator jets to elements $v_1, \dots, v_s$ of the *actual* kernel K_i , then varies them by elements of $\mathfrak{m}^N K_i$. Choose generators $w_1, \dots, w_t$ of $\mathfrak{m}^N K_i$ and set

$$v_j(c) = v_j + \sum_{\ell=1}^t c_{\ell j} w_\ell. \quad (3.4)$$

At every geometric point off the origin, $\mathfrak{m}^N$ is the unit ideal. Consequently the parameters in (3.4) map onto the space of $r \times s$ fiber matrices, where $r = ar_i$ and $s = a\beta_i$. The rank-deficient locus has codimension $s - r + 1$. By (3.3),

$$s - r + 1 > d.$$

The incidence of points and parameters where generation fails therefore has dimension strictly smaller than the parameter space. Its image has proper Zariski closure. Over an infinite field a suitable parameter choice generates everywhere off the origin; Nakayama’s lemma handles the origin.

Crucially, the perturbations occur inside actual kernels. The argument does not truncate arbitrary differential matrices and hope that their products stay zero. Exactness is built into the successive surjections onto these kernels. Subsequent comparison changes act as target changes on already constructed maps and preserve their marked column depths.

The original prescribed-jet note explicitly describes this as a prescribed-jet form of the usual generator dimension count and supplies the incidence proof. It does not invoke an unidentified stronger relative Forster–Swan theorem.

3.3. The last syzygy and the determinant obstruction. At the final syzygy the surplus in (3.3) disappears. Instead, the final kernel is locally free everywhere, hence finite projective. Quillen–Suslin makes it free [8, 9]. The contemporary notes used [2, Theorem 6] as a checked exposition of this input.

There remains a genuine basis-jet obstruction. If $U \in \mathrm{GL}_r(A)$ describes a desired completed basis and $u = \det U$, an invertible polynomial matrix cannot have an arbitrary determinant jet: its determinant must be constant. Rescale the first basis vector by $u(0)/u$. This preserves its column order and makes the determinant constant. After a constant diagonal normalization, the resulting jet lies in

$$\mathrm{SL}_r(S/\mathfrak{m}^N).$$

Over this local Artinian ring it is a product of elementary matrices; lifting their coefficients to S gives the required polynomial basis. For $r = 1$, the normalized assertion is immediate.

For finite fields, the construction is first made over an algebraic closure. Its finitely many polynomial coefficients descend to a finite extension k'/k . Restricting scalars then gives the required direct sum over k , multiplying all relevant counts by $[k' : k]$.

4. DOCUMENTED CHRONOLOGY

The recorded bottleneck was the graded-to-local transfer. The periodic-box note explicitly said that an exact global polynomial resolution was still required and that arbitrary formal complexes had not been algebraized. Passing to associated graded objects could lose exactness.

This supplied a concrete reason to seek the construction above, rather than a general wish to simplify power-series matrices.

The following times are September 4, 2026, Pacific daylight time. They come from public progress messages, source-search timestamps, and file creation or update metadata. They mark recorded events, not necessarily the first instant at which a thought occurred.

Time	Recorded event
16:02:21	The coordinating agent searched the exact scopes of Forster–Swan and Quillen–Suslin.
16:10:46	The intersection agent reported that the intermediate generator counts and final projective syzygy worked.
16:11:27	The filtrations agent identified preservation of selected column jets as the part not settled by the all-deep argument.
16:12:41	It reported generic kernel perturbations and final determinant normalization.
16:12:47	The all-deep globalization note was created.
16:16:07	The prescribed-jet note was created; it was finalized at 16:22:50.

The all-deep note attributes the proposal to the coordinating agent and its independent check and write-up to the intersection agent. The prescribed-jet note credits the filtrations agent and records that the coordinating agent also independently proposed the incidence argument and determinant normalization. These are contemporaneous internal attributions, not evidence of priority over existing literature.

4.1. Relation to the overall duration. The first complete Lech candidate was saved at 16:25:41 that day. The earlier chronology places this about 37 hours 54 minutes after the structural prompt on September 3 at 02:31:31, and about 54 hours 23 minutes after the first Lech session on September 2 at 10:03:02. These are elapsed wall-clock intervals, not cumulative agent-hours. The short globalization window should not be represented as the time taken to find or check the whole argument. The preceding investigation had already supplied the mathematical setting and exposed the failed transfer routes.

5. PUBLISHED ANTECEDENTS AND THE LIMITS OF ATTRIBUTION

5.1. Sources demonstrably used at the time. The original all-deep note cites [6, Theorem 1.1] for the exact Forster–Swan bound and its field scope, and [2, Theorem 6] for Quillen–Suslin. It also cites the Stacks Project for completion and finite projectivity. The generator-theory source points back to [4, Corollary 3] and [5, Corollary 5]. The manuscript bibliography later includes the original Forster, Swan, Quillen, and Suslin papers.

There is no external citation in those notes for the complete amplified prescribed-column proposition. The archival record supports “constructed in this investigation from the cited classical tools,” not the stronger claim “first discovered here.”

5.2. Related literature identified during this review. A particularly relevant older paper is Eisenbud’s *Adic approximation of complexes, and multiplicities* [3]. Its introduction explicitly credits Peskine–Szpiro, Section 1.6, together with Artin approximation, for descending finite-projective-dimension modules from completions to finite étale extensions of suitable local rings. This is a genuine antecedent to the general theme. Its stated descent destination and approximation result are different from the exact global polynomial, Betti-preserving, prescribed-column construction above. It should not be cited as that proposition without a further argument.

Another nearby example is De Stefani–Polstra–Yao, *Global Frobenius Betti numbers and Frobenius Euler characteristics*, Section 4 [1]. They use Forster–Swan on syzygies to compare local and global generator counts. This reinforces the established methodological ancestry of the global generator approach. It has not been verified here as a source of the precise prescribed-column statement, or as a paper used in its historical development.

These two references were identified as related work in the present review. They should not retroactively be described as the source from which the earlier construction was obtained. Nor does failure to find an exact match in a targeted search justify a novelty claim.

5.3. Suggested attribution in the detailed manuscript. The following would accurately credit the known ingredients:

We construct the required globalization by applying Forster–Swan successively to the syzygies of a sufficiently large direct sum, and Quillen–Suslin to the final projective syzygy. Preservation of selected column orders follows from a relative generator argument, proved below.

This avoids an unsupported claim in either direction about whether the assembled construction has appeared before.

6. WHAT CAN BE SAID ABOUT THE PROMPT AND THE ORIGIN OF THE IDEAS

The available evidence is a record of mathematical dependencies, source consultations, saved arguments, and public progress reports. It cannot identify a particular training example or certify that an apparently new combination was independent of all prior literature. An account of “where the idea came from” should therefore distinguish documented source use from a retrospective mathematical explanation.

The broader prompt asked for sustained structural work instead of further finite case analysis, and for continued investigation and checking. The research record documents that change of direction. I found no instruction supplying the direct-sum generator surplus, the kernel-perturbation argument, or the determinant normalization. The prompt’s identifiable contribution was to the process and allocation of effort, rather than an explicit suggestion of this construction.

Whether the same construction would have emerged without that prompt is not determined by the record. There is no controlled comparison supporting “it would have happened anyway.” Likewise, the present expert circulation and internal checks should not be represented as completed external validation of the whole Lech manuscript.

APPENDIX A. WORKSPACE EVIDENCE

The following paths are relative to `C:/Users/agerko/Documents/Lech/`. These are the mathematical records supporting this report; public session messages and tool timestamps supply the finer chronology.

- (1) `output/paper/lech_equal_characteristic.tex`, lines 838–1003: prescribed generators, basis jets, and the proposition on globalization with selected columns.
- (2) `forster_swan_globalization_of_deep_local_resolutions_20260904.md`, Sections 1–4: the all-deep construction, intermediate generator counts, final projective syzygy, and consulted sources.
- (3) `prescribed_jet_globalization_of_sparse_deep_local_resolutions_20260904.md`, Sections 1–4: relative generation, determinant normalization, selected-column globalization, and finite-field descent.

- (4) `periodic_degree_boxes_finite_length_graded_repair_and_boundary_moment_20260904.md`, lines 314–323: the exact polynomial-resolution interface still required by the periodic-box construction.
- (5) `structural_root_audit_20260903.md`, Sections 576 and 578: acceptance of the all-deep step and then the prescribed-jet refinement.
- (6) `RESEARCH.md` and `candidate_equal_characteristic_lech_proof_20260904.md`: the broader investigation chronology and first assembled candidate.

REFERENCES

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